

Apply the integral test to decide the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{n^a}$$

where  $a \in \mathbb{R}$   
is some fixed  
parameter.

Assume  $a > 0$

The function  $f(t)$

$$f(t) = \frac{1}{t^a}$$

is a decreasing positive function.

$$\frac{1}{t^a} \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

By the integral test,  $\sum_{n=1}^{\infty} \frac{1}{n^a}$

converges iff  $\int_1^{\infty} \frac{1}{t^a} dt$  converges/  
is finite.

Case  $a=1$  is special

$$\begin{aligned} \int_1^{\infty} \frac{1}{t} dt &= \left[ \ln(t) \right]_1^{\infty} \\ &= \lim_{T \rightarrow \infty} \ln(T) \end{aligned}$$



So when  $a > 1$   $\int_1^{\infty} \frac{1}{t^a} dt$  is finite.

So  $\sum_{n=1}^{\infty} \frac{1}{n^a}$  converges also, when  $a > 1$ .

eg.  $\sum_{n=1}^{\infty} \frac{1}{n^2}$ ,  $\sum_{n=1}^{\infty} \frac{1}{n^3}$ , ... are

convergent series.

The family of series  $\sum_{n=1}^{\infty} \frac{1}{n^a}$

are called hyper harmonic series.

which are divergent when  $a \leq 1$   
and convergent when  $a > 1$ .

---

Q 6 (i).

Recall the geometric series test:

The geometric series  $\sum_{n=1}^{\infty} ar^{n-1}$

with initial term  $a$  and common ratio  $r$

For  $a \neq 0$  this converges iff  $|r| < 1$





